

Design & Analysis of Algorithm

Pre requisits:

- ✓ → programming language (C/C++)
- ✓ → Data structures

Course Outline

1. Algorithm Analysis → O, θ, Ω

Analyse
Complexity
of the
problem

- Time Complexity / Space Complexity
- Best Case / Avg Case / worst case

2. Algorithm Design

- Different techniques to solve the problem. "Backtracking, Branch & Bound"
- Divide & Conquer, Greedy, Dynamic,

* What is an algorithm?

- Finite set of "Deterministic" instructions to perform a task.

(- There is always a unique well defined step to follow for every given

step and input combination.

→ If this condition is not satisfied

↳ Then the algorithm is
"Non-Deterministic"

* Name 3 algorithms to sort 'n' element, and give their time complexity.

✓ → Quick — $O(n \log n)$

✓ → Merge — $O(n \log n)$

selection —

Bubble sort — $O(n^2)$

Heap

"The problem of sorting cannot be solved in time less than $O(n \log n)$ using comparison based algo"

"Searching problem cannot be solved in time less than $O(\log n)$ "

Text Book:

- Horowitz & Sahani — (3)
- Cormen et. al. — (1)
- Brassard & Bratley — (2)